

MMD Aggregated Two-Sample Test

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Abstract

We propose two novel nonparametric two-sample kernel tests based on the Maximum Mean Discrepancy (MMD). First, for a fixed kernel, we construct an MMD test using either permutations or a wild bootstrap, two popular numerical procedures to determine the test threshold. We prove that this test controls the probability of type I error non-asymptotically. Hence, it can be used reliably even in settings with small sample sizes as it remains well-calibrated, which differs from previous MMD tests which only guarantee correct test level asymptotically. When the difference in densities lies in a Sobolev ball, we prove minimax optimality of our MMD test with a specific kernel depending on the smoothness parameter of the Sobolev ball. In practice, this parameter is unknown and, hence, the optimal MMD test with this particular kernel cannot be used. To overcome this issue, we construct an aggregated test, called MMDAgg, which is adaptive to the smoothness parameter. The test power is maximised over the collection of kernels used, without requiring held-out data for kernel selection (which results in a loss of test power), or arbitrary kernel choices such as the median heuristic. We prove that MMDAgg still controls the level non-asymptotically, and achieves the minimax rate over Sobolev balls, up to an iterated logarithmic term. Our guarantees are not restricted to a specific type of kernel, but hold for any product of one-dimensional translation invariant characteristic kernels. We provide a user-friendly parameter-free implementation of MMDAgg using an adaptive collection of bandwidths. We demonstrate that MMDAgg significantly outperforms alternative state-of-the-art MMD-based two-sample tests on synthetic data satisfying the Sobolev smoothness assumption, and that, on real-world image data, MMDAgg closely matches the power of tests leveraging the use of models such as neural networks.

Keywords: two-sample testing, kernel methods, minimax adaptivity

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1 Introduction

We consider the problem of nonparametric two-sample testing, where we are given two independent sets of i.i.d. samples, and we want to determine whether these two samples come from the same distribution. This fundamental problem has a long history in statistics and machine learning, with numerous real-world applications in various fields, including clinical laboratory science (Miles et al., 2004), genomics (Chen and Qin, 2010), biology (Fisher et al., 2006), geology (Vermeesch, 2013) and finance (Horváth et al., 2013).

To compare samples from two probability distributions, we use a statistical test of the null hypothesis that the two distributions are equal, against the alternative hypothesis that they are different. Many such tests exist, and rely on different assumptions. If we assume that the two probability distributions are Gaussian with the same variance, then we can perform a Student’s t-test (Student, 1908) to decide whether or not to reject the null hypothesis. However, the t-test is parametric in nature, and designed only for comparing two Gaussian distributions. By contrast, our interest is in nonparametric tests, which are sensitive to general alternatives, without relying on specific distributional assumptions. An example of such a nonparametric test is the Kolmogorov–Smirnov test (Massey Jr, 1951) which uses as its test statistic the largest distance between empirical distribution functions of the two samples. The limitation of the Kolmogorov–Smirnov test, however, is that it applies only to univariate data, and its multivariate extension is challenging (Bickel, 1969).

The test statistic we consider is an estimate of the Maximum Mean Discrepancy (MMD—Gretton et al., 2007, 2012a) which is a kernel-based metric on the space of probability distributions. The MMD is an integral probability metric (Müller, 1997) and hence is defined as the supremum, taken over a class of smooth functions, of the difference of their expectations under the two probability distributions. This function class is taken to be the unit ball of a characteristic Reproducing Kernel Hilbert Space (Aronszajn, 1950; Fukumizu et al., 2008; Sriperumbudur et al., 2011), so the Maximum Mean Discrepancy depends on the choice of kernel. We work with a wide range of kernels, each parametrised by their bandwidths.

There exist several heuristics to choose the kernel bandwidths. In the Gaussian kernel case, for example, bandwidths are often simply set to the median distance between pairs of points from both samples (Gretton et al., 2012a). This strategy for bandwidth choice